

Smart AI Notes Generator

Project Report

My Profile

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About Me: I am pursuing my Degree in Data Science and Applications at the Indian Institute of Technology Madras, where I am developing a strong foundation in technology and data-driven problem solving. I have a keen interest in web application development, ML, AI and enjoy creating applications that blend analytics, learning, and seamless user experience. I am driven by curiosity and a desire to build impactful, AI Based real-world digital solutions.

Smart Notes Generator

AI-Powered Study Notes and Educational Content Generation System

Abstract

Smart Notes Generator is a full-stack AI-powered educational web application designed to help students, self-learners, and educators organize complex knowledge into structured study notes automatically. The application uses Google Gemini AI to analyze educational topics, classify them intelligently, generate detailed learning content, and export professional PDF study materials.

The system combines a Vue.js frontend with a Flask backend to create a responsive and interactive educational platform. Users can explore any topic, select relevant subtopics, generate highly detailed AI-created notes, view them in an interactive slide-based interface, and download professionally formatted PDF documents.

The project demonstrates practical integration of modern AI technologies with web application development and automated document generation.

Introduction

Traditional note-making and study material preparation require significant time and effort. Students often struggle with:

- Organizing complex topics
- Identifying important subtopics
- Creating concise summaries
- Generating structured revision notes
- Understanding coding-related subjects deeply
- Exporting notes into printable formats

Smart Notes Generator solves these problems using artificial intelligence.

The application acts like a virtual tutor capable of:

- Organizing topics logically
 - Generating structured educational content
 - Detecting whether a topic is code-related
 - Producing code examples automatically
 - Creating summaries and flash notes
 - Generating practice questions
 - Exporting professional PDF documents
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Objectives

The main objectives of the project are:

1. Build a modern AI-powered study notes platform
2. Automatically organize educational topics
3. Generate highly detailed study material
4. Detect programming-related topics intelligently
5. Generate code examples when required
6. Create interactive educational slides
7. Enable professional PDF exports
8. Build a scalable frontend-backend architecture
9. Use REST APIs for communication
10. Integrate Google Gemini AI effectively

Features

- AI topic organization
 - AI-generated educational slides
 - Intelligent code-topic detection
 - Code examples and explanations
 - Interactive slide viewer
 - Syntax-highlighted code blocks
 - Flash notes and summaries
 - PDF export system
 - Vue.js frontend
 - Flask backend
 - Gemini AI integration
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Technologies Used

Frontend Technologies

Technology	Purpose
HTML5	Page structure
CSS3	Styling and responsive design
JavaScript	Application logic
Vue.js 3 (CDN)	Reactive frontend framework
Highlight.js	Syntax highlighting for code

Backend Technologies

Technology	Purpose
Python	Backend programming language
Flask	Web framework
Google Gemini API	AI content generation
ReportLab	PDF generation
Logging Module	Debugging and monitoring

Project Structure

```
smart-notes/  
├── app.py  
├── requirements.txt  
├── static/  
│   ├── style.css  
│   └── app.js  
└── templates/  
    └── index.html
```

System Architecture

Frontend Layer

The frontend is developed using Vue.js without a build system.

Responsibilities:

- User interaction
- State management
- Rendering educational slides
- Sending API requests
- PDF download handling
- Dynamic transitions and animations

Backend Layer

The Flask backend acts as the processing engine.

Responsibilities:

- Handling API requests
 - Managing Gemini AI calls
 - Generating educational content
 - Processing topic classification
 - Creating PDF documents
 - Error handling and logging
-

AI Layer

Google Gemini AI is used for:

- Topic organization
 - Topic classification
 - Detailed note generation
 - Code example generation
 - Summary generation
 - Flash note creation
 - Practice question generation
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Application Workflow

Step 1 – Topic Search

The user enters a topic such as:

- Machine Learning
- Quantum Physics
- Python Programming

The frontend sends the topic to:

POST /api/topics/explore

Step 2 – Topic Organization

Gemini AI analyzes the topic and organizes it into:

- Topic groups
- Subtopics
- Descriptions

The frontend displays selectable topic cards.

Step 3 – Topic Selection

The user selects relevant subtopics.

Examples:

- Neural Networks
- Supervised Learning
- Regression

Step 4 – AI Topic Classification

Before generating notes, the backend asks Gemini:

Is this topic code-related?

This intelligent AI-based classification replaces hardcoded keyword detection.

Step 5 – Notes Generation

The frontend sends selected topics to:

POST /api/notes/generate

Gemini generates:

- Overview slides
- Detailed explanations
- Code examples
- Summaries
- Flash notes
- Practice questions

Step 6 – Slide Viewer

Generated notes appear inside an interactive slide viewer.

Features include:

- Slide navigation
- Animated transitions
- Syntax-highlighted code blocks
- Summary sections
- Flash note grids

Step 7 – PDF Export

The frontend sends notes data to:

POST /api/notes/download-pdf

The backend generates a professional A4 PDF using ReportLab.

API Endpoints

1. Explore Topics API

Endpoint

POST /api/topics/explore

Purpose

Organizes a topic into educational groups and subtopics.

Request Example

```
{
  "topic": "Machine Learning"
}
```

Response Example

```
{
  "mainTopic": "Machine Learning",
  "topicGroups": []
}
```

2. Generate Notes API

Endpoint

POST /api/notes/generate

Purpose

Generates detailed educational slides.

Features

- AI topic classification
 - Code-aware educational generation
 - Practice question generation
 - Summary creation
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3. Download PDF API

Endpoint

POST /api/notes/download-pdf

Purpose

Exports notes as downloadable PDF documents.

PDF Generation System

The PDF engine is built using ReportLab.

Features:

- A4 page format
- Professional typography
- Slide-based page layout
- Code block rendering
- Styled summaries
- Flash note grids
- Headers and footers

The system dynamically generates PDFs from AI-generated JSON data.

Frontend Design

The frontend follows a clean educational interface.

Color Palette

Color	Usage
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#1e2a4a	Dark navy primary
#6366f1	Indigo accents
#f59e0b	Flash notes
#f7f8fc	Background

Key Features

AI-Powered Topic Organization

Automatically structures educational topics.

Intelligent Topic Classification

Uses AI to detect programming-related topics.

Code-Aware Learning System

Generates:

- code examples
- syntax explanations
- debugging tips
- best practices

Interactive Slide Viewer

Displays educational content dynamically.

PDF Export System

Creates printable study material.

Syntax Highlighting

Improves code readability.

Advantages of the System

1. Saves time for students

2. Generates organized study material
3. Supports coding education
4. Creates printable notes instantly
5. Highly scalable architecture
6. AI-driven educational assistance
7. Beginner-friendly interface

Future Improvements

Future enhancements may include:

- User authentication
- Database integration
- Multi-language support
- Audio summaries
- Interactive quizzes
- Dark mode
- Cloud storage
- Collaborative note sharing
- Real-time editing
- AI voice tutor integration

Conclusion

Smart Notes Generator demonstrates the practical implementation of AI in education technology. By combining Flask, Vue.js, Gemini AI, and ReportLab, the application creates a complete intelligent learning platform capable of organizing knowledge, generating educational content, and exporting professional study material.

The project successfully integrates modern web technologies with generative AI to improve the educational experience and reduce the manual effort involved in note creation.